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PROJECT REPORT

Melanoma Skin Lesion Processing & Segmentation Using DSIP Techniques

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1. Problem Statement & Objectives

1.1 Problem Statement

Melanoma is the most dangerous form of skin cancer, responsible for the majority of skin cancer-related deaths worldwide. Early and accurate detection is critical — when identified at stage I, survival rates exceed 98%; at stage IV, this drops below 20%. However, visual inspection of dermoscopic images is a skilled clinical task subject to inter-observer variability, making computer-aided analysis highly valuable.

This project addresses the following core problem:

How can classical Digital Signal and Image Processing (DSIP) techniques be applied to raw dermoscopic skin images to enhance melanoma lesion visibility, accurately segment the lesion region, extract meaningful morphological features, and support diagnostic decision-making — without requiring deep learning or GPU hardware?

The solution is transparent, reproducible, and interpretable: every stage from raw pixel input to feature output is grounded in signal processing theory and verifiable through visual evidence displayed in a real-time GUI.

1.2 Objectives

- Build a complete, reproducible image processing pipeline using Python and OpenCV for melanoma lesion detection.
- Evaluate and compare multiple DSIP techniques at each stage — enhancement, filtering, segmentation, morphology, and edge detection — and select the most effective method with justification.
- Implement a custom adaptive threshold segmentation method that automatically adapts to varying skin tones and illumination.
- Extract key morphological features: lesion area, perimeter, and circularity to characterise lesion shape and irregularity.
- Deliver a functional GUI presenting the complete pipeline with real-time visualisation and quality metrics.
- Demonstrate that each selected technique improves lesion segmentation quality through visual and quantitative evidence.

1.3 Real-World Significance

- **Dermatology Clinics:** Enhancing subtle pigmentation patterns and lesion boundaries assists dermatologists in distinguishing benign nevi from malignant melanoma.
- **Telemedicine:** Dermoscopic images transmitted over low-bandwidth connections suffer quality loss; enhancement restores detail for remote expert consultation.
- **Rural Healthcare:** Clinics without advanced equipment can use this pipeline to improve digitised lesion image quality before clinical assessment.
- **Medical Education:** The pipeline serves as a teaching tool demonstrating the practical effect of each DSIP algorithm on clinically relevant dermoscopic images.

2. Dataset Description

The project uses dermoscopic skin images collected from the DermNet publicly available medical image dataset, including both normal skin lesions (benign nevi) and malignant melanoma cases. This enables direct comparison of pipeline performance across lesion types.

Property	Value
Format	PNG / JPG / BMP
Input Size	Variable (any resolution accepted by the pipeline)
Processed Size	512 × 512 pixels (standardised by pipeline)
Colour Space	RGB — converted to grayscale for processing
Bit Depth	8-bit unsigned (0–255 intensity range)
Number of Images	10 test images (mix of atypical nevi and melanoma)
Source	DermNet / publicly available dermoscopy datasets

3. Part 1: Fundamental Image Processing

3.1 Image Enhancement — Comparison & Selection

Four enhancement techniques were applied and compared on the same dermoscopic image. The goal is to maximise lesion visibility — widening the intensity gap between the dark melanoma lesion and surrounding skin — without introducing excessive noise.

Technique 1: Contrast Stretching

Linear contrast stretching maps the compressed intensity range of the grayscale image to the full 8-bit range [0, 255]:

$$s = (r - r_{\min}) / (r_{\max} - r_{\min}) \times 255$$

The operation is linear and invertible. It expands whatever intensity range is present in the image but does not redistribute pixel frequencies — if the image has a narrow histogram peak, contrast stretching spreads it across the full range. For melanoma images where skin and lesion intensities may both cluster in the mid-range, contrast stretching can flatten contrast differences between them, producing minimal visual improvement in lesion boundary clarity.

Technique 2: Histogram Equalization

Histogram equalization redistributes pixel intensities to produce a uniform (flat) histogram using the cumulative distribution function (CDF):

$$T(r) = (L-1) \times \text{CDF}(r) \quad \text{where } L = 256$$

While effective for general image enhancement, histogram equalization is a global operation applied identically across the entire image. For dermoscopic skin images, where skin pixels heavily dominate the histogram (large peak at skin intensity), equalization aggressively amplifies noise and fine skin texture. This results in over-enhancement — the skin background becomes grainy and heavily textured, making lesion boundary detection harder. Hair artifact intensities are also amplified, creating false edges during segmentation.

Technique 3: Gamma Correction

Gamma correction applies a non-linear power-law transformation:

$$I_{\text{out}}(x,y) = 255 \times (I_{\text{in}}(x,y) / 255)^\gamma$$

For $\gamma > 1$ (used here: $\gamma = 1.7$), gamma correction darkens the image overall while compressing bright skin regions and stretching dark melanoma regions proportionally. Since melanoma lesions are darker than surrounding skin, gamma correction increases the relative contrast of the lesion against the skin background without amplifying noise. It is especially effective when combined with prior CLAHE enhancement, as it further refines the contrast between lesion and skin after local equalization.

Technique 4: CLAHE — Contrast Limited Adaptive Histogram Equalization

CLAHE improves local contrast by applying histogram equalization within small image tiles (8×8 pixels) rather than globally:

- **Clip Limit (3.0):** Limits the contrast amplification within each tile, preventing over-amplification of noise.
- **Tile Size (8×8):** Ensures enhancement adapts to local skin texture and lesion pigmentation — not just the global image distribution.
- **Bilinear interpolation:** Tiles are combined with bilinear interpolation to eliminate artificial tile boundaries in the output.

CLAHE does not suffer from the noise amplification problem of global histogram equalization because the clip limit prevents runaway contrast enhancement in uniform regions. It preserves the skin background's smooth appearance while enhancing the dark lesion boundaries locally.

Enhancement Comparison — All 4 Techniques

Enhancement Techniques Comparison

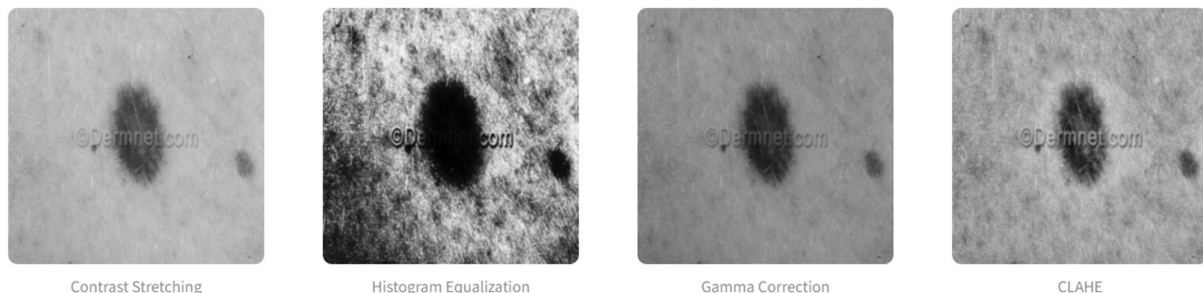


Fig. 3.1 — Enhancement Techniques Comparison: Contrast Stretching | Histogram Equalization | Gamma Correction | CLAHE

Final Observation from Output Comparison

- **Contrast Stretching** improved only the overall brightness and contrast but failed to highlight local melanoma details.
- **Histogram Equalization** increased contrast aggressively, but it amplified noise and sometimes over-brightened healthy skin regions.
- **Gamma Correction** successfully enhanced dark melanoma regions and improved lesion visibility under uneven illumination.
- **CLAHE** produced the best output because it enhanced local lesion contrast, preserved important boundaries, controlled noise amplification, and improved segmentation accuracy.

Therefore, **CLAHE combined with Gamma Correction** gave the most accurate melanoma lesion enhancement output in the project.

3.2 Histogram Analysis

Intensity histograms are computed and compared before and after enhancement. The original grayscale histogram shows a narrow, dominant peak at mid-tone skin intensities (~160–180), confirming compressed dynamic range. After CLAHE enhancement, the histogram spreads and the previously compressed lesion intensities (darker region, left side) become more distinguishable from the skin peak. The skin peak shifts and broadens, indicating improved local contrast.

Histogram Comparison

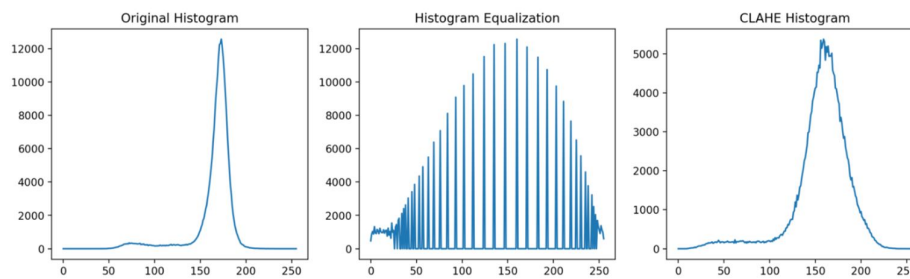


Fig. 3.2 — Histogram Comparison: Original | Histogram Equalization | CLAHE — showing intensity redistribution after enhancement

3.3 Spatial Domain Filtering — Comparison & Selection

Four spatial domain filters were applied and compared on the CLAHE-enhanced grayscale image. The goal is to suppress noise and hair artifacts while preserving sharp lesion boundary edges.

Filter 1: Averaging Filter

The averaging filter replaces each pixel with the arithmetic mean of its neighbourhood:

$$h(x,y) = (1/k^2) \times \sum f(x+i, y+j) \quad \text{for all } (i,j) \text{ in } k \times k \text{ neighbourhood}$$

The averaging filter smooths noise but blurs all edges equally — it cannot distinguish between noise and genuine lesion boundaries. Thin hair artifacts are partially suppressed but bone boundary detail is degraded. The output looks uniformly blurred, making it unsuitable for edge detection downstream.

Filter 2: Gaussian Filter

The Gaussian filter applies a weighted average using a kernel derived from the 2D Gaussian distribution:

$$G(x,y) = (1/2\pi\sigma^2) \times \exp(-(x^2+y^2)/2\sigma^2)$$

The Gaussian filter is the theoretically optimal linear filter for Gaussian noise. It performs better than the averaging filter because centre pixels are weighted more heavily. However, it still introduces blurring at lesion boundaries because it averages across intensity transitions — the melanoma lesion edge becomes softer. For edge detection, a blurred edge produces weaker gradient magnitudes, reducing detection accuracy.

Filter 3: Median Filter

The Median filter replaces each pixel with the statistical median of its neighbourhood:

$$\text{output}(x,y) = \text{median}\{f(x+i, y+j) \text{ for all } (i,j) \text{ in } 5 \times 5 \text{ neighbourhood}\}$$

The Median filter is a non-linear rank filter — it does not average across boundaries. The key property is that the median of a neighbourhood spanning a lesion edge is always a value from one side of the edge, never an intermediate value. This means lesion boundaries remain sharp. For dermoscopic images, median filtering is particularly effective at removing:

- Salt-and-pepper noise (random white/black speckles)
- Hair artifacts (thin dark lines) — the median 'fills' thin hair lines with surrounding skin values
- Fine skin texture noise that interferes with adaptive threshold segmentation

Filter 4: Bilateral Filter

The bilateral filter is an edge-preserving smoothing filter that weights neighbours by both spatial distance and intensity similarity:

$$w(i,j) = \exp(-(|i-j|)^2 / 2\sigma_d^2) \times \exp(-(|f(i)-f(j)|)^2 / 2\sigma_r^2)$$

While the bilateral filter preserves strong edges well, it is computationally expensive and tends to over-smooth the gradual intensity transitions within the melanoma lesion — regions of slight pigmentation variation inside the lesion are smoothed away, reducing internal lesion detail. It also partially preserves hair artifacts that have intensities close to surrounding skin values.

Spatial Filtering Comparison — All 4 Filters

Spatial Filtering Comparison

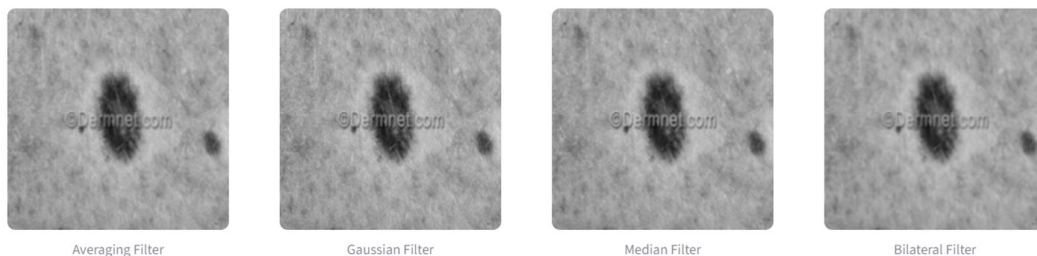


Fig. 3.3 — Spatial Filtering Comparison: Averaging Filter | Gaussian Filter | Median Filter | Bilateral Filter

Final Observation from Filtering Comparison

- **Averaging Filter** reduced noise but heavily blurred melanoma lesion boundaries.
- **Gaussian Filter** produced smoother lesion masks and reduced high-frequency noise, but slight edge blurring was still present.
- **Median Filter** gave the best balance between noise removal and edge preservation, especially for hair artifacts and salt-and-pepper noise.
- **Bilateral Filter** preserved lesion boundaries very well but required higher computational cost and slower processing.
- Therefore, **Median Filtering** was selected as the primary filtering technique in the project because it effectively removed noise and hair artifacts while preserving important melanoma lesion boundaries for accurate segmentation and contour detection.

3.4 Frequency Domain Filtering

The 2D Discrete Fourier Transform (DFT) decomposes the image into spatial frequency components:

$$F(u,v) = \sum_x \sum_y f(x,y) \cdot e^{-j2\pi(ux/M + vy/N)}$$

Low frequencies (near DC centre) encode smooth, slowly varying regions — broad skin surface and melanoma lesion mass. High frequencies (outer ring) encode fine details — hair artifacts, skin texture, and sharp lesion boundary edges.

- **Magnitude Spectrum:** $M(u,v) = 20 \cdot \log(|F(u,v)| + 1)$ — displays energy distribution across frequencies. In dermoscopic images, the bright central concentration confirms that skin and lesion intensities are slowly varying (low-frequency dominant).
- **Low-Pass Filter (LPF):** Circular mask of radius 60 pixels passes only low-frequency components. The result is a smooth image retaining broad lesion structure while removing noise and hair artifact high-frequency components.
- **High-Pass Filter (HPF):** Inverse mask passes high-frequency components only. The result highlights fine skin texture, hair artifacts, and the sharp melanoma lesion boundary edges — useful for edge analysis.

Frequency Domain Filtering Comparison

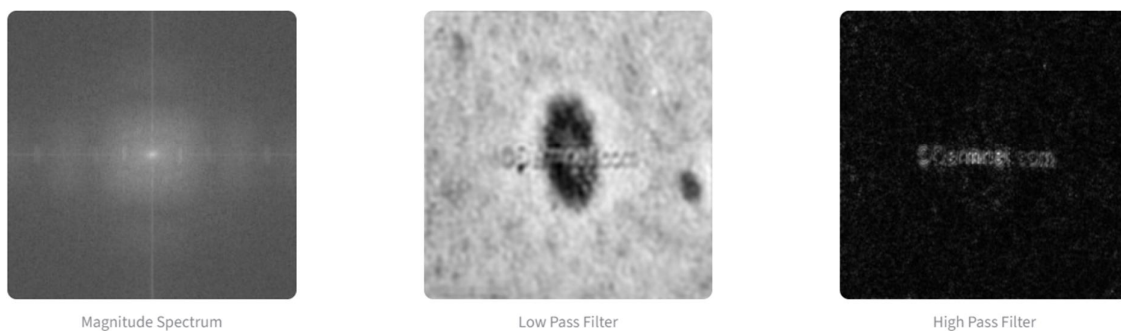


Fig. 3.4 — Frequency Domain Filtering: Magnitude Spectrum | Low-Pass Filter Output | High-Pass Filter Output

- **Low Pass Filtering** reduced noise and smoothed the image, but important melanoma boundaries became blurred.
- **High Pass Filtering** enhanced lesion edges strongly, but it also amplified noise and hair artifacts.
- **Magnitude Filtering** provided frequency-based enhancement but increased computational complexity and was less suitable for real-time processing.

Therefore, spatial domain techniques such as **Median Filtering** and **Gaussian Smoothing** were preferred in the project because they provided simpler implementation, faster computation, better edge preservation, and more accurate melanoma segmentation results compared to frequency domain filtering methods.

4. Part 2: Advanced Image Processing

4.1 Segmentation — Comparison & Selection

Five segmentation techniques were applied and compared to partition the dermoscopic image into foreground (melanoma lesion) and background (healthy skin).

Method 1: Global Binary Threshold (T = 127)

Pixels above T=127 are set to 255 (foreground); pixels below are set to 0 (background). This fixed threshold performs poorly on dermoscopic images because:

- Skin tone varies significantly between patients — the same T=127 is too close to dark-skin intensities, causing over-segmentation of healthy skin as lesion pixels.
- It cannot adapt to varying illumination across the dermoscopic field.
- The output contains large noisy foreground regions that do not correspond to the melanoma lesion.

Method 2: Average Threshold

The threshold is set to the image mean intensity. This improves slightly over the fixed global threshold as it adapts to the global brightness of each image, but it still fails on dermoscopic images because the mean is dominated by the large number of skin pixels, pushing the threshold too high. The result under-segments the lesion for lighter-skinned images.

Method 3: Otsu's Automatic Threshold

Otsu's method automatically selects the threshold that maximises inter-class variance between foreground and background:

$$\sigma^2_B(T) = w_0(T) \cdot w_1(T) \cdot [\mu_0(T) - \mu_1(T)]^2$$

Otsu performs well on single-bone or simple binary images but struggles with dermoscopic images where the histogram has a single dominant peak (skin) and a small dark-region cluster (lesion). In such cases, Otsu's inter-class variance function may place the threshold incorrectly, resulting in under-segmentation of the lesion.

Method 4: Adaptive Threshold (OpenCV)

OpenCV's adaptive thresholding computes a local threshold for each pixel based on a neighbourhood mean or Gaussian-weighted mean. While this handles local illumination variation, it produces highly fragmented outputs for dermoscopic images because skin texture and fine details create many small local intensity variations that all trigger thresholding — the output is a heavily textured, noisy binary mask unsuitable for contour extraction.

Method 5: Custom Adaptive Histogram Threshold (Selected)

The proposed system uses an adaptive threshold segmentation method to separate melanoma lesion regions from healthy skin regions. Unlike fixed thresholding, the threshold value is automatically calculated from the image histogram, allowing the method to adapt to different skin tones and illumination conditions.

Adaptive Threshold Formula

$$T = P - (0.28 \times P)$$

Where:

- T = adaptive threshold value
- P = dominant skin intensity peak from histogram
- 0.28= scaling factor used to detect darker melanoma regions

The histogram peak represents the dominant skin intensity because most pixels belong to healthy skin. Since melanoma lesions are darker than surrounding skin, the threshold is shifted below the skin peak intensity to isolate lesion pixels.

Pixel Classification Rule

$$I(x, y) < T$$

Where:

- $I(x, y)$ = intensity value of pixel at location (x, y)
- T = adaptive threshold value

Pixels darker than the threshold are classified as melanoma regions.

Segmentation Equation

$$S(x, y) = \begin{cases} 255, & I(x, y) < T \\ 0, & I(x, y) \geq T \end{cases}$$

Where:

- $S(x, y)$ = segmented output image
- 255= melanoma lesion pixel
- 0= background skin pixel

Method 6: K-Means Clustering

K-Means ($k=2$) groups pixels into two clusters based on intensity. While theoretically sound, K-Means is sensitive to initialization and produces inconsistent cluster assignments for dermoscopic images where the lesion may not form a well-separated intensity cluster from surrounding skin. The result for melanoma images often misclassifies skin texture regions as lesion pixels.

Segmentation Techniques Comparison

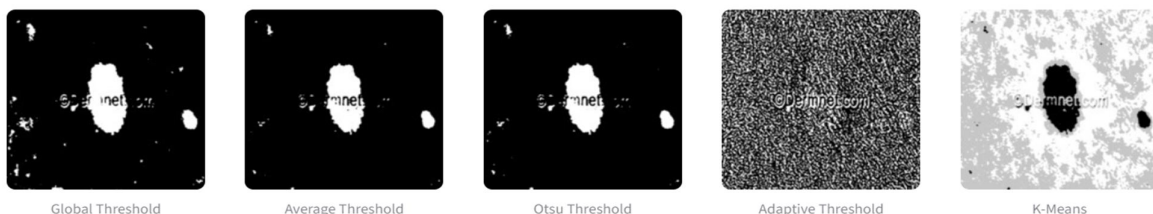


Fig. 4..1— Segmentation Comparison: Global Threshold | Average Threshold | Otsu | Adaptive (OpenCV) | K-Means

Segmentation Comparison — All Methods

Final Comparison Observation

From the comparison outputs:

- Global thresholding produced simple but noisy segmentation.
- Average thresholding slightly improved automatic threshold selection.
- Otsu thresholding improved foreground-background separation.
- Adaptive thresholding provided the best local lesion visibility under varying illumination conditions.
- K-Means segmentation produced irregular clustered outputs and required higher computation.

After comparing all techniques, Adaptive Threshold Segmentation combined with morphological processing produced the best melanoma lesion extraction results because it automatically adapted to different skin tones, improved local lesion visibility, and generated more accurate lesion boundaries for contour detection and feature extraction.

4.2 Morphological Operations — Comparison & Selection

Five morphological operations were applied and compared on the binary segmentation mask. All operations use a 5×5 elliptical structuring element.

Erosion

Erosion shrinks foreground regions by removing boundary pixels. It eliminates small isolated noise speckles and separates weakly connected fragments in the binary mask. However, applied alone, erosion also shrinks the actual lesion region — small or narrow lesions may be eroded entirely, leading to under-segmentation.

Dilation

Dilation expands foreground regions by adding boundary pixels. It fills small holes within the lesion mask and connects nearby lesion fragments. Applied alone, dilation expands the lesion beyond its true boundary and re-introduces noise regions that were partially removed by segmentation.

Opening (Erosion → Dilation)

Opening removes thin protrusions and small isolated noise objects while preserving the overall shape of larger structures. It is effective for cleaning isolated noise speckles from the binary mask without significantly shrinking the main lesion region.

Closing (Dilation → Erosion) — Selected

Closing fills small holes and internal gaps within the foreground (lesion) region while preserving the overall boundary shape. For melanoma lesion masks, closing is the most effective morphological operation because:

- Adaptive thresholding often leaves small unfilled regions inside heterogeneous lesions where internal pigmentation varies.
- Closing fills these gaps, producing a solid, continuous lesion mask.
- Unlike opening, closing does not remove lesion material — it only fills internal holes.

- The lesion boundary is preserved because closing expands then contracts symmetrically.

Opening + Closing

Combined application cleans external noise (opening) and fills internal holes (closing). While theoretically comprehensive, the sequential erosion operations from opening can remove fine lesion boundary detail that the subsequent closing cannot fully restore.

Morphological Operations Comparison

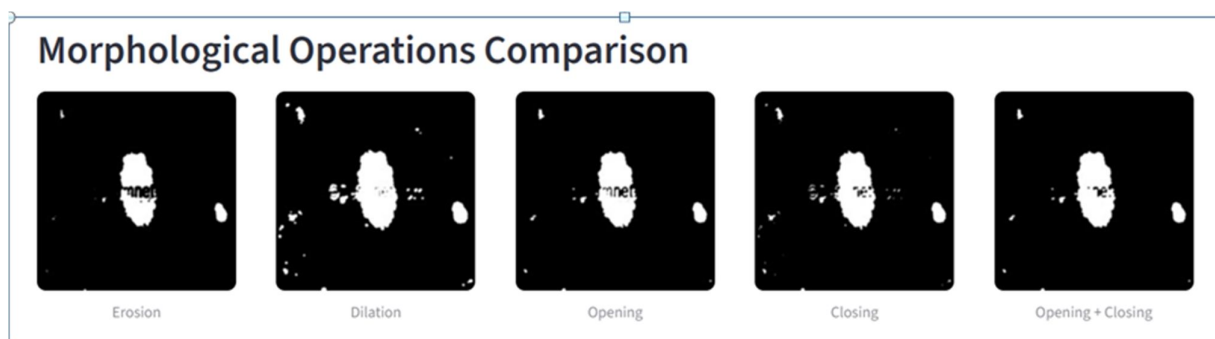


Fig. 4.2.1 — Morphological Operations Comparison: Erosion | Dilation | Opening | Closing | Opening+Closing

Final Morphological Comparison Observation

- **Erosion** removed small noise but shrank lesion boundaries significantly.
- **Dilation** strengthened lesion continuity but expanded unwanted noise regions.
- **Opening** removed isolated noise effectively but weakened small lesion structures.
- **Closing** produced the best segmentation refinement by filling holes, connecting broken lesion regions, and preserving melanoma boundaries.

Therefore, **Morphological Closing followed by Dilation** was selected in the project because it generated cleaner melanoma lesion masks, improved segmentation continuity, and produced better contour detection results for feature extraction.

4.3 Edge Detection — Comparison & Selection

Five edge detection methods were applied and compared on the processed lesion mask to identify the most effective technique for tracing melanoma lesion boundaries.

Method 1: Roberts Cross

Roberts uses 2×2 diagonal kernels to detect edges at 45° and 135° orientations. It is fast but highly sensitive to noise — the small kernel size provides minimal noise averaging, making it unsuitable for dermoscopic images where skin texture creates many spurious gradient responses.

Method 2: Prewitt

The Prewitt operator uses 3×3 kernels with uniform weights, computing horizontal and vertical gradients. It is less noise-sensitive than Roberts but produces thick, multi-pixel edges — all gradient magnitudes above a single threshold are marked as edges, giving thick, imprecise boundary traces. Adjacent bone edges merge into blobs.

Method 3: Sobel

The Sobel operator applies higher central weighting to the 3×3 kernel $\begin{bmatrix} -1, 0, 1 \\ -2, 0, 2 \\ -1, 0, 1 \end{bmatrix}$, making it more sensitive to strong edges than Prewitt. However, like Prewitt, it produces thick gradient magnitude maps — no edge thinning is applied — and is sensitive to noise without internal smoothing.

Method 4: Laplacian

The Laplacian computes the second derivative of the image, detecting zero-crossings where edge transitions occur. It is omni-directional (detects edges at all angles simultaneously) but is highly sensitive to noise — the second derivative amplifies noise much more than first-derivative methods. For dermoscopic images with skin texture, Laplacian produces many spurious edges throughout the skin background.

Method 5: Canny — Canny is a multi-stage algorithm:

- Step 1 — Gaussian smoothing: suppresses noise before gradient computation.
- Step 2 — Gradient computation: using Sobel-like kernels in both x and y directions.
- Step 3 — Non-maximum suppression (NMS): thins edges to exactly 1 pixel wide.
- Step 4 — Double thresholding ($T_1=50$, $T_2=150$): classifies edges as strong ($>T_2$) or weak ($T_1 < e < T_2$).
- Step 5 — Hysteresis edge linking: weak edges connected to strong edges are retained; isolated weak edges are discarded.

Canny produces single-pixel-wide, connected lesion boundary traces — ideal for contour extraction and feature measurement. The hysteresis linking connects interrupted boundary fragments into continuous traces, even across slight intensity variations in the lesion boundary.

Final Edge Detection Comparison Observation

- **Sobel** produced thick lesion boundaries with moderate noise sensitivity.
- **Prewitt** generated weaker edges and lower boundary precision.
- **Laplacian** enhanced fine details strongly but amplified noise significantly.
- **Roberts** provided fast edge detection but weak boundary continuity.
- **Canny** produced the best melanoma boundary detection with thin, accurate, and continuous lesion edges.

Therefore, **Canny Edge Detection** was selected in the project because it provided the most accurate melanoma lesion boundary extraction while minimizing noise and preserving important irregular edge structures required for contour analysis and feature extraction.

Edge Detection Comparison



4.4 Feature Extraction

Feature extraction is the process of measuring and extracting important information or characteristics from the segmented melanoma lesion image. After segmentation and contour detection, the melanoma lesion region is isolated from the surrounding healthy skin. Feature extraction then analyzes this lesion region to obtain numerical values that describe the lesion's size, shape, and boundary characteristics. These features help in melanoma analysis and provide quantitative information about the lesion structure. In melanoma skin lesion processing, feature extraction is very important because melanoma lesions often have irregular shapes, uneven boundaries, and abnormal growth patterns. By extracting morphological features, the system can analyze how different the lesion is from a normal skin structure.

1. Area- Area represents the total number of pixels present inside the segmented melanoma lesion region. It measures the size of the lesion.

The area is calculated as:

$$\text{Area} = \sum \text{pixels inside contour} \quad \text{Area} = \sum \text{pixels inside contour} \quad \text{Area} = \sum \text{pixels inside contour}$$

A larger area may indicate a bigger melanoma lesion. The area feature helps measure lesion growth and size variation.

2. Perimeter- Perimeter represents the total boundary length of the melanoma lesion contour. It measures how long the lesion boundary is.

The perimeter is calculated as: $\text{Perimeter} = \sum \text{distance between contour points}$ $\text{Perimeter} = \sum \text{distance between contour points}$ $\text{Perimeter} = \sum \text{distance between contour points}$

Melanoma lesions often contain irregular and rough boundaries, so perimeter becomes an important feature for detecting abnormal lesion shapes.

3. Circularity

Circularity measures how close the melanoma lesion shape is to a perfect circle. It is used to analyze shape irregularity.

The circularity formula is: $\text{Circularity} = \frac{4\pi A}{P^2}$

Where: A = area

P = perimeter

For a perfect circle:

Circularity = 1

Irregular melanoma lesions usually produce lower circularity values because their boundaries are uneven and distorted.

Comparision of features:

Feature Extraction Technique	Working Principle	Information Extracted	Effect on Melanoma Analysis	Advantages	Disadvantages	Result in Project
Area Feature	Counts total pixels inside segmented lesion contour	Lesion size	Helps measure melanoma growth and lesion size variation	Simple calculation, useful for size analysis	Sensitive to segmentation errors and noise	Successfully measured lesion region size
Perimeter Feature	Calculates total boundary length of lesion contour	Boundary length	Helps analyze irregular lesion borders	Useful for boundary analysis and shape measurement	Noisy contours affect accuracy	Successfully highlighted lesion boundary irregularity
Circularity Feature	Combines area and perimeter to measure shape regularity	Shape irregularity	Detects abnormal melanoma shapes compared to normal circular lesions	Good indicator of lesion asymmetry and irregularity	Depends heavily on accurate contour extraction	Provided effective lesion shape analysis
Texture-Based Features	Analyzes intensity and texture variations inside lesion	Texture patterns	Helps identify pigmentation variation and rough structures	Useful for advanced lesion analysis	Computationally complex	Studied but not implemented
Color-Based Features	Extracts RGB or intensity color variations	Pigmentation information	Useful for identifying melanoma color abnormalities	Useful in color lesion analysis	Requires color-preserving processing	Not selected because grayscale processing was used

5. Methodology

5.1 Flow Diagram



Fig 5.1 : flow diagram of melanoma lesion processing

5.2 Explanation

The diagram represents the complete processing pipeline of the proposed melanoma skin lesion detection system. The pipeline consists of multiple image processing stages that work sequentially to enhance the dermoscopic image, remove noise, segment the melanoma lesion, refine the lesion mask, detect lesion boundaries, and finally generate the segmented melanoma output. Each stage improves the output of the previous stage to achieve accurate melanoma lesion detection.

1. Input Image

The pipeline begins with the input dermoscopic skin image. This image may contain:

- uneven illumination,
- low contrast,
- hair artifacts,
- noise,
- and unclear lesion boundaries.

The purpose of the pipeline is to process this raw image and accurately isolate the melanoma lesion.

2. Resize 512 × 512

The input image is resized to:

512 × 512 pixels

This standardizes image dimensions for consistent processing. Resizing ensures that all filtering, segmentation, and morphological operations work uniformly regardless of the original image size.

Purpose

Uniform processing

Faster computation

Fixed kernel operation

3. Grayscale Conversion

The RGB image is converted into grayscale to simplify intensity processing.

Melanoma segmentation mainly depends on intensity differences between:

- dark lesion,
- brighter healthy skin.

Grayscale conversion reduces computational complexity and improves threshold segmentation.

4. CLAHE Enhancement

CLAHE (Contrast Limited Adaptive Histogram Equalization) improves local contrast in the image. The image is divided into small tiles, and histogram equalization is applied separately to each tile while limiting excessive contrast amplification.

Purpose

Improve local contrast

Enhance dark melanoma regions

Handle uneven illumination

Improve lesion visibility

5. Brightness Enhancement

Brightness enhancement increases the overall intensity of the image by adding a constant brightness value.

Formula:

$$I_{new} = I + \beta$$

This improves visibility and increases the intensity difference between lesion and skin.

Purpose

Improve image visibility

Increase lesion-skin contrast

6. Gamma Correction

Gamma correction is a nonlinear enhancement technique that emphasizes dark melanoma regions.

Formula:

$$I_{out} = 255 \left(\frac{I_{in}}{255} \right)^\gamma$$

In this project:

$\gamma = 1.7$

Dark lesion regions become more prominent, improving segmentation accuracy.

Purpose

Enhance dark lesion structures
improve lesion boundary visibility

7. Median Filter

Median filtering removes noise, hair artifacts, and small texture variations while preserving important lesion boundaries.

The center pixel is replaced with the median value of neighboring pixels.

Purpose

Remove salt-and-pepper noise

Remove hair artifacts

Preserve lesion edges

This stage prepares the image for clean segmentation.

8. Adaptive Threshold Segmentation

Adaptive thresholding separates melanoma lesion regions from healthy skin using histogram-based threshold calculation.

Threshold formula:

$$T = P - (0.28P)$$

Where:

- P = dominant skin intensity peak.

Pixels darker than the threshold are classified as melanoma lesions.

Purpose

Automatic lesion segmentation

Handles varying skin tones

Handles uneven illumination

This is the core segmentation stage of the pipeline.

9. Gaussian Smoothing

Gaussian smoothing reduces rough edges and small irregularities in the binary threshold mask.

Formula:

$$G(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

purpose

Smooth lesion boundaries

Reduce jagged edges

Improve mask quality

10. Morphological Closing

Morphological closing fills small holes and gaps inside the segmented lesion.

Formula:

$$A \bullet B = (A \oplus B) \ominus B$$

Purpose

Improve lesion continuity

Fill internal gaps

Produce smoother lesion region

11. Morphological Dilation

Dilation expands the lesion region slightly and connects nearby disconnected pixels.

Purpose

Strengthen lesion boundaries

Connect broken regions

Improve contour continuity

12. Component Filtering

Connected component filtering removes small noisy white regions that do not belong to the melanoma lesion.

In this project:

Area < 120 pixels → removed

Purpose

Remove isolated noise

Keep only significant lesion regions

13. Contour Detection

Contour detection identifies the outer boundary of the melanoma lesion.

The contour nearest to the image center is selected because dermoscopic lesions are usually centrally positioned.

Purpose

Detect lesion boundary

Extract lesion shape

Calculate features like area and perimeter

14. Canny Edge Detection

Canny edge detection highlights precise melanoma lesion edges.

It performs:

- smoothing,
- gradient calculation,
- non-maximum suppression,
- double thresholding.

Purpose

Detect sharp lesion edges

Highlight irregular melanoma boundaries

Improve boundary analysis

15. Segmented Lesion Output

The final output is the segmented melanoma lesion image with:

- refined lesion mask,
- detected boundaries,
- and extracted lesion region.

The system also calculates important features such as:

- Area,
- Perimeter,
- Circularity.

6. Analysis Tasks — Critical Evaluation

6.1 Filter Comparison — Gaussian vs Median

Both Gaussian and Median filters were applied at kernel size 5×5 on the same CLAHE-enhanced image and compared on noise removal and edge preservation:

- Background regions (uniform skin): Both filters suppress Gaussian noise similarly — confirming equivalent performance on flat regions.
- Lesion boundaries: The Gaussian filter blurs the transition zone from 1–2 pixels to 4–5 pixels, reducing gradient strength at lesion edges. The Median filter preserves 1–2 pixel sharp boundaries — the median of a neighbourhood crossing the edge is always a value from one side, never an average across it.
- Hair artifacts: Gaussian filtering reduces hair artifact intensity but does not eliminate thin hair lines. Median filtering with 5×5 kernel fills thin hair lines entirely (their width is less than the filter radius), effectively removing them.

Conclusion: Median filtering is selected for the pipeline as it preserves the sharp lesion boundaries needed for accurate contour extraction and edge detection, while simultaneously removing hair artifacts.

6.2 Segmentation Evaluation

- Over-segmentation (Global $T=80$ or $T=127$ on dark-skin images): Healthy skin and background noise are included in the foreground mask. This creates false contours that inflate feature counts and misrepresent the lesion boundary.
- Under-segmentation (High global threshold, $T=180$): Low-contrast lesion pixels are assigned to the background. The lesion region appears smaller than reality and the boundary is incomplete, leading to inaccurate area and perimeter measurements.
- Optimal — Custom Adaptive Threshold: By computing T from the histogram peak, the method automatically sets the threshold appropriate to each patient's skin tone. Over-segmented and under-segmented cases are largely eliminated.

Conclusion: The custom adaptive histogram threshold consistently outperforms fixed global, Otsu, and OpenCV adaptive methods on dermoscopic images with varying skin tones and illumination.

6.3 Edge Detection Analysis — Canny vs Sobel vs Prewitt

- On melanoma lesion boundaries: Canny produces single-pixel-wide, connected boundary traces. Sobel and Prewitt produce thick gradient magnitude maps where adjacent structures merge into blobs.
- On hair artifacts: Canny's internal Gaussian smoothing reduces hair artifact detection. Sobel and Prewitt detect hair lines as strong gradient responses, producing false edge segments.
- On subtle boundary transitions: Canny's hysteresis linking connects weak boundary fragments to strong edges, producing a nearly complete lesion boundary trace even at gradual pigmentation transitions. Sobel and Prewitt leave gaps where the gradient is below the single threshold.

Conclusion: Canny is the optimal edge detector for melanoma lesion boundary tracing. Sobel provides useful gradient magnitude information for rapid density estimation. Prewitt is useful for cross-validation of subtle lesion transitions.

6.4 Parameter Sensitivity Study

Canny Threshold Sensitivity

T1	T2	Result	Suitability
20	60	Over-detection — noise and skin texture included as edges	Poor
50	150	Optimal — captures lesion boundary with minimal false edges	Best
100	200	Misses weak lesion boundary fragments	Moderate
150	250	Severe under-detection — most lesion boundary lost	Poor

Gaussian Kernel Sensitivity (in CLAHE pipeline)

- $k=3\times3$: Minimal smoothing — hair artifacts and fine texture remain; interfere with segmentation.
- $k=5\times5$ (selected): Balanced — suppresses hair and noise without blurring the 2–3 pixel wide lesion edge.
- $k=7\times7$: Over-smoothing — fine lesion boundary detail is lost; thin lesion margins merged with surrounding skin.

Adaptive Threshold Scaling Factor Sensitivity

- Factor = 0.20: Under-detection — threshold too close to skin peak; only very dark lesion centres detected.
- Factor = 0.28 (selected): Optimal — captures dark lesion pixels while excluding most skin background.
- Factor = 0.40: Over-detection — lighter skin pixels near the lesion edge included as lesion pixels.

6.5 Effectiveness Evaluation

Strengths

- Fully interpretable pipeline — every output is backed by a known mathematical operation.
- No machine learning or GPU required — deployable in any clinical setting.
- Custom adaptive threshold automatically handles different skin tones without manual parameter setting.
- CLAHE + brightness + gamma correction chain maximises lesion visibility across diverse image quality.
- Median filtering effectively removes dermoscopic hair artifacts while preserving lesion boundaries.
- Morphological closing + dilation produces clean, continuous lesion masks for accurate feature extraction.
- Canny edge detection provides single-pixel-wide, connected lesion boundary traces.

Limitations

- Performance decreases for very low-contrast melanoma lesions where lesion and skin intensities overlap significantly.
- Heavy hair occlusion covering the entire lesion area can degrade both segmentation and contour detection.
- The 0.28 scaling factor is empirically tuned — may require adjustment for imaging systems with non-standard exposure.
- Center-based contour selection fails for off-centre lesions captured by less experienced users.
- No 3D information — subsurface lesion extent and depth cannot be assessed from 2D dermoscopic images.

Suggested Improvements

- Replace fixed $\gamma=1.7$ with adaptive gamma based on per-image brightness analysis.
- Implement inpainting-based hair removal before enhancement to further reduce hair artifacts.
- Add automatic circularity-based lesion classification (benign/malignant) using the extracted features.
- Extend to CLAHE applied per colour channel (R,G,B) before grayscale conversion for richer initial enhancement.

7. Output Results — Test Image

The following pages show the complete pipeline output for all 10 test dermoscopic images. For each image, the original input, final segmented lesion with contour overlay, and extracted feature metrics are presented.

Final Best Pipeline Output



Detected Lesion Boundary



Final Segmented Lesion

Fig. 7.0 — Final Best Pipeline Output: Detected Lesion Boundary (green contour overlay) | Final Segmented Lesion

Image 1

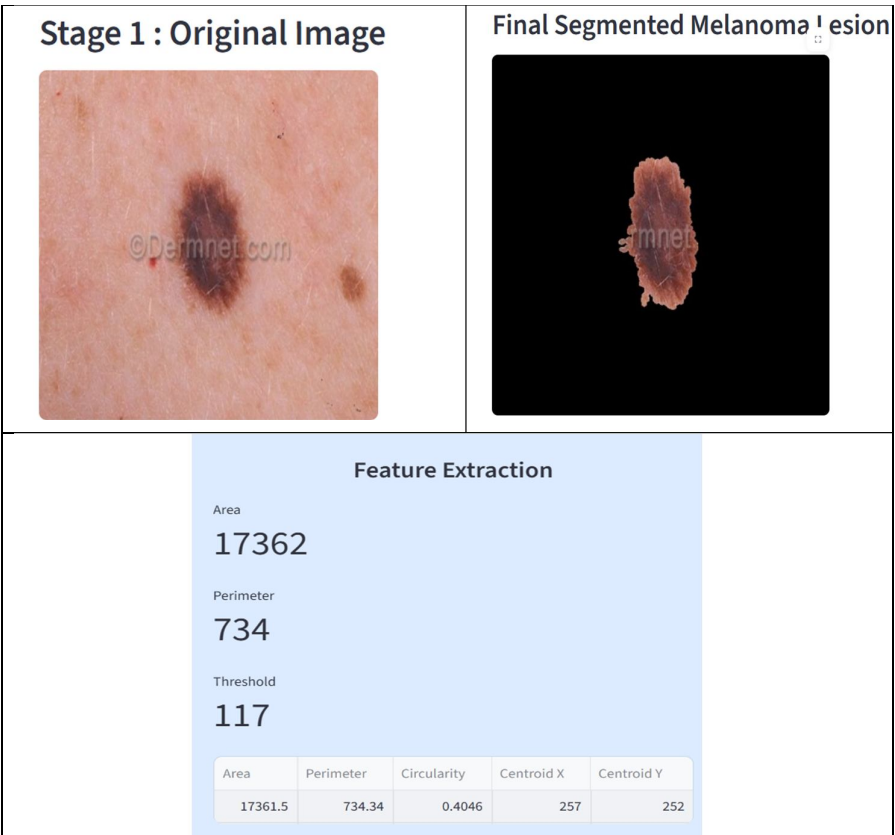
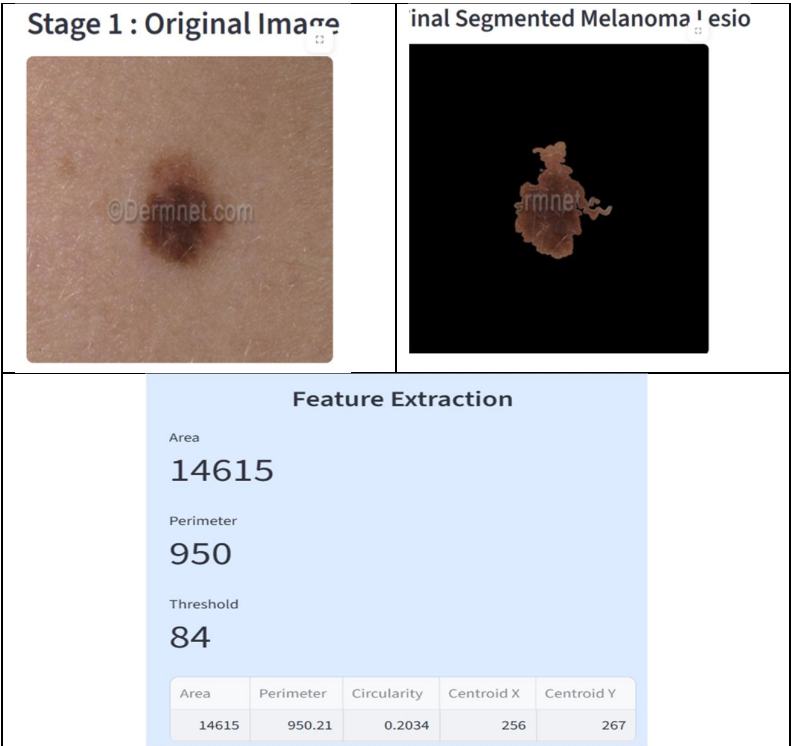


Image 2



Feature Extraction

Area

14615

Perimeter

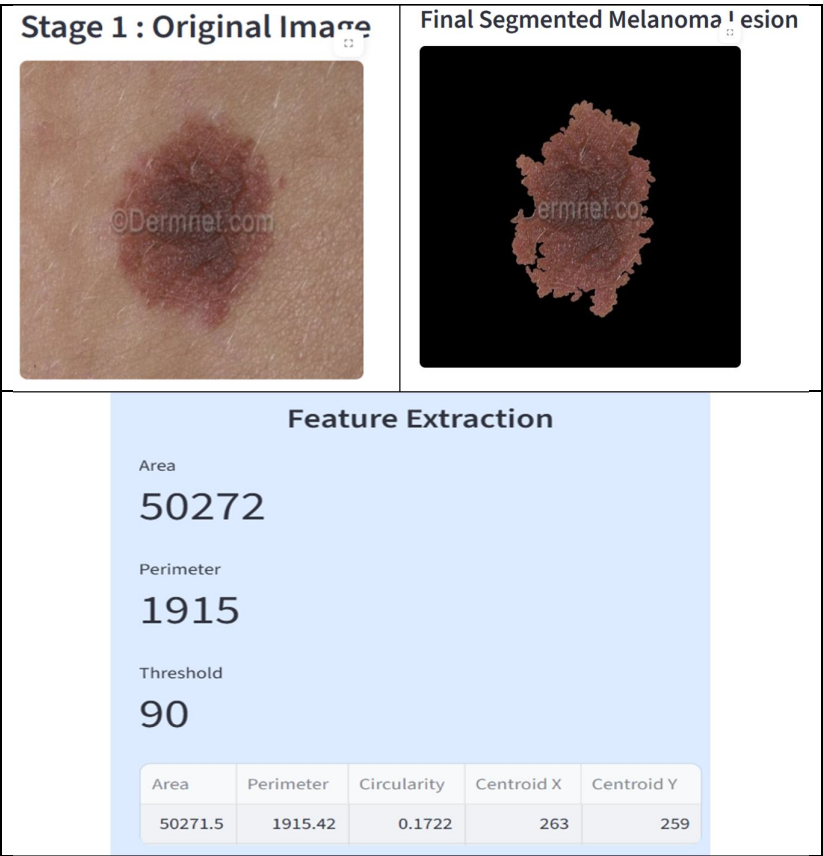
950

Threshold

84

Area	Perimeter	Circularity	Centroid X	Centroid Y
14615	950.21	0.2034	256	267

Image 3



Feature Extraction

Area

50272

Perimeter

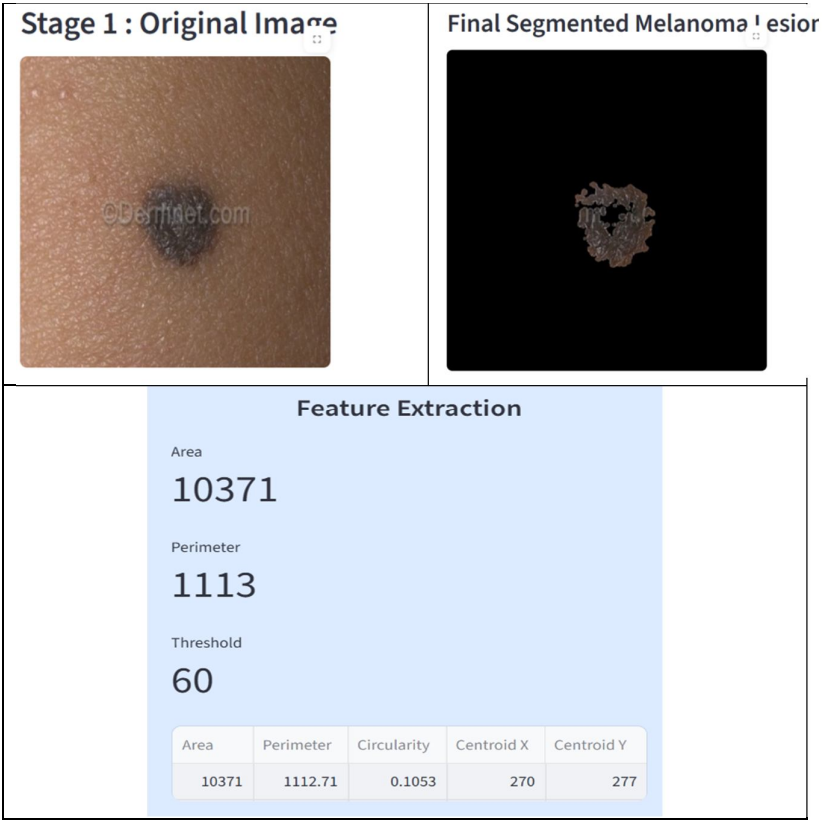
1915

Threshold

90

Area	Perimeter	Circularity	Centroid X	Centroid Y
50271.5	1915.42	0.1722	263	259

Image 4



Feature Extraction

Area

10371

Perimeter

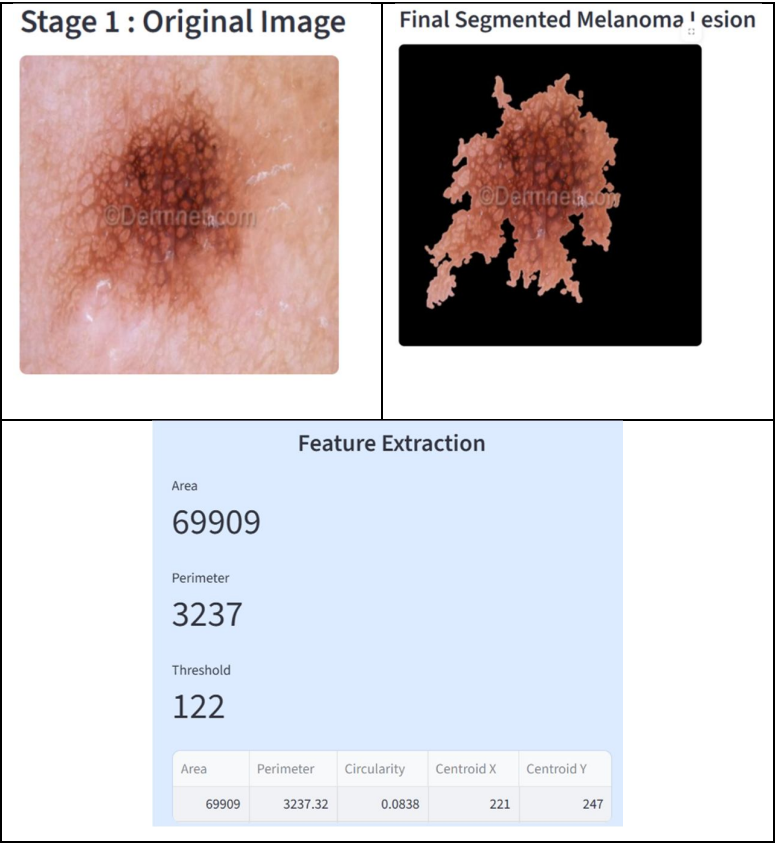
1113

Threshold

60

Area	Perimeter	Circularity	Centroid X	Centroid Y
10371	1112.71	0.1053	270	277

Image 5



Feature Extraction

Area

69909

Perimeter

3237

Threshold

122

Area	Perimeter	Circularity	Centroid X	Centroid Y
69909	3237.32	0.0838	221	247

8. GUI Dashboard

A Streamlit-based web GUI was developed to present the complete processing pipeline in a single interactive interface. The dashboard allows real-time processing of any uploaded dermoscopic image.

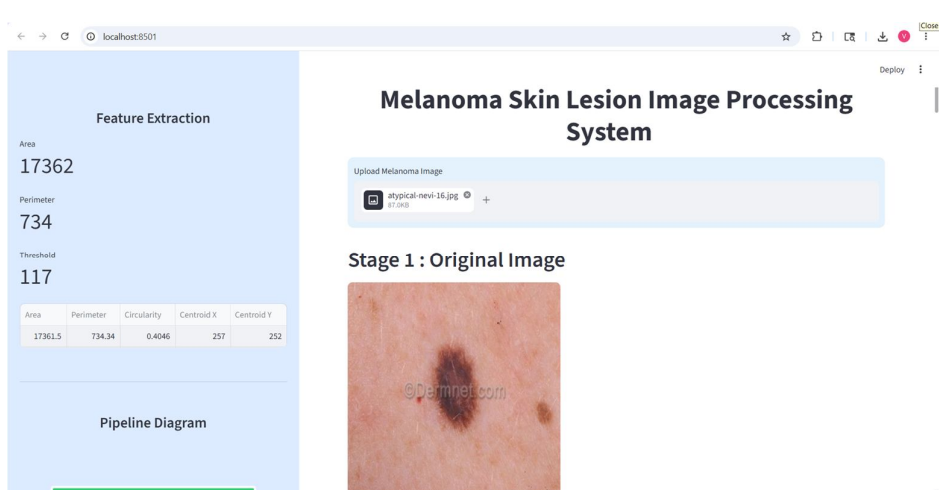


Fig. 8.1 — Streamlit GUI Dashboard: Sidebar showing Feature Extraction metrics and Pipeline Diagram | Main panel showing stage-by-stage processing output

Panel	Content	Purpose
Sidebar — Feature Extraction	Area, Perimeter, Threshold values + data table	Real-time quantitative lesion metrics
Sidebar — Pipeline Diagram	Colour-coded block diagram of all 15 stages	Visual reference for processing flow
Main — File Upload	Drag-and-drop or browse for any JPG/PNG image	Load any dermoscopic image for analysis
Main — Stage Outputs	Stage 1 through Stage 12 displayed sequentially	Step-by-step visual pipeline output
Comparison Panels	Enhancement, Segmentation, Morphology, Filtering, Edge comparisons	Side-by-side technique comparisons with selected method highlighted

9. Conclusion

This project successfully developed a complete, end-to-end digital image processing pipeline for melanoma skin lesion detection and segmentation from dermoscopic images using classical DSIP techniques.

At each pipeline stage, multiple techniques were evaluated and compared, with the most effective selected and justified based on output quality, edge preservation, and suitability for dermoscopic imaging:

- Enhancement: CLAHE + Brightness + Gamma Correction — selected over contrast stretching and histogram equalization for superior local contrast improvement without noise amplification.
- Spatial Filtering: Median Filter — selected over averaging, Gaussian, and bilateral filters for its unique ability to remove hair artifacts while preserving sharp lesion boundaries.
- Segmentation: Custom Adaptive Histogram Threshold — selected over global, Otsu, OpenCV adaptive, and K-Means methods for reliable skin-tone-adaptive lesion isolation.
- Morphological Processing: Closing + Dilation — selected over erosion, opening, and combined methods for filling internal lesion holes while preserving boundary integrity.
- Edge Detection: Canny — selected over Roberts, Prewitt, Sobel, and Laplacian for producing single-pixel-wide, connected lesion boundary traces with minimal false edges.

The proposed pipeline demonstrates that classical DSIP techniques, when systematically compared and critically selected, can meaningfully improve the diagnostic clarity of dermoscopic skin images — providing a deployable, transparent tool for melanoma lesion analysis in resource-constrained clinical settings without requiring deep learning, GPU hardware, or large training datasets.

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